| | |
|---|---|
| | Using $EPE = \frac{1}{2} kx$<br>Additional $EPE = \frac{1}{2} k(x_{final}^2 - x_{initial}^2) = \frac{1}{2}(100)(0.040^2 - 0.030^2) = 0.035$ J |
| **9** | **C**<br><br>Total gain in KE = loss in GPE of Y – loss of GPE of X – work done by friction<br>$= (5.0)(9.81)(2.0) - (4.0)(9.81)(2.0 \sin 30) - 5(2)$<br>$= 48.86 \approx 49$ J |
| **10** | **A**<br>Two forces, tension and weight, act on the pendulum. The resultant force should not |